

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHE 511 — Past Questions & Materials
by Renance

Course: CHE 511
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 500

This study pack was compiled and restructured by Renance for CHE 511 students at Federal University of Petroleum Resources, Effurun. It brings together 2 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 2 — CHE 511 LINEAR PROGRAMMING PRACTICE QUESTIONS

Level 500 • First Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

LINEAR PROGRAMMING

THE GRAPHICAL METHOD

- ① Use the graphical method to solve the following LP problem.

$$\text{Maximize } Z = 15x_1 + 10x_2$$

Subject to the constraints

$$\textcircled{1} 4x_1 + 6x_2 \leq 360$$

$$x_1, x_2 \geq 0$$

$$\textcircled{2} 3x_1 + 8x_2 \leq 180$$

$$\textcircled{3} 0x_1 + 5x_2 \leq 200$$

Solution

Algebraically in mathematical form

x_1 - horizontal axis

x_2 - vertical axis

$$\text{Treat } \textcircled{1} \text{ as } 4x_1 + 6x_2 = 360$$

The easiest way to plot this line is to find any two points that satisfy the equation and then to draw a straight line through them. The two points are generally the points at which the line intersects the x_1 & x_2 axes
eg when $x_1 = 0$ $6x_2 = 360$, $x_2 = 60$ Similarly
when $x_2 = 0$ $4x_1 = 360$, $x_1 = 90$

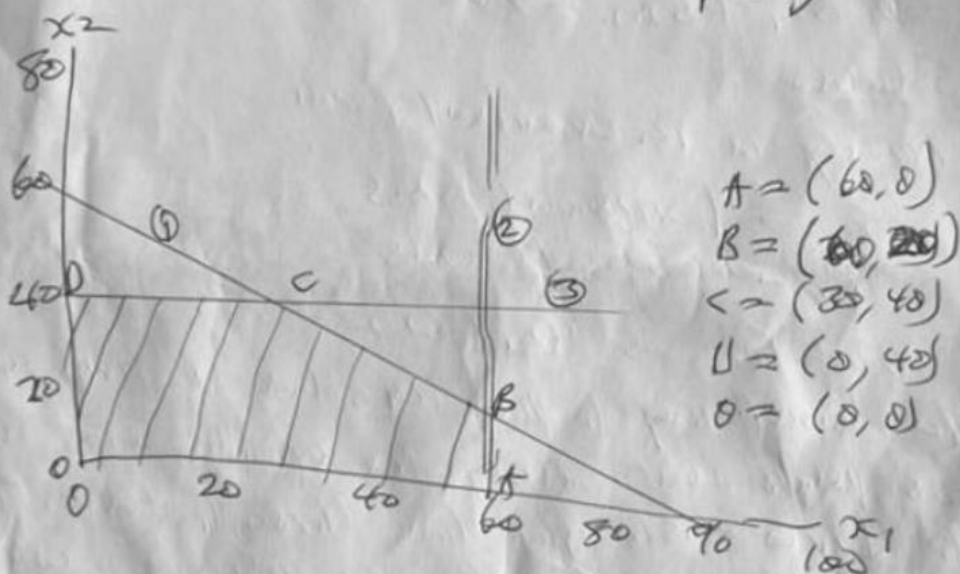


The 2 pt are connected by a straight line. Any point above the constraint line violates the inequality condition. But any point below the line does not

violate the constraint. The inequality and non-negative constraints can only be satisfied by the shaded area (Feasible) region.

Similarly $3x_1 = 180 \Rightarrow x_1 = 60$
 $5x_2 = 200, x_2 = 40$

Since all constraint lines be plotted, the area which is bounded by all the constraints lines including all the boundary point is called the feasible region (solution space)



Extreme point	Co-ordinate	Objective function Value $Z = 15x_1 + 10x_2$
O	(0, 0)	$15(0) + 10(0) = 0$
A	(60, 0)	$60(15) + 10(0) = 900$
B	(60, 20)	$15(60) + 10(20) = 1100$
C	(30, 40)	$15(30) + 10(40) = 850$
D	(0, 40)	$15(0) + 10(40) = 400$

$Z \rightarrow \text{maximum} = 1100$, at (60, 20)
 $Z = 1100, x_1 = 60, x_2 = 20$

② The ABC company has been a producer of picture tubes for television sets and certain printed circuits for radio. The company has just expanded into full scale production and marketing of AM and AM-FM radios. It has built a new plant that can operate 48 hours per week. Production of AM radio in the new plant will require 2 hrs and production of an AM-FM radio will require 3 hours. Each AM radio will contribute Rs 40 to profit while AM-FM radio contribute Rs 80 to profit. The marketing department after extensive research has determined that a maximum of 15 AM radio and 10 AM-FM radios can be sold each week.

④ Formulate a linear programming model to determine the optimum production mix of AM and FM radios that will maximize profit.

⑤ Solve the above using the graphic method.

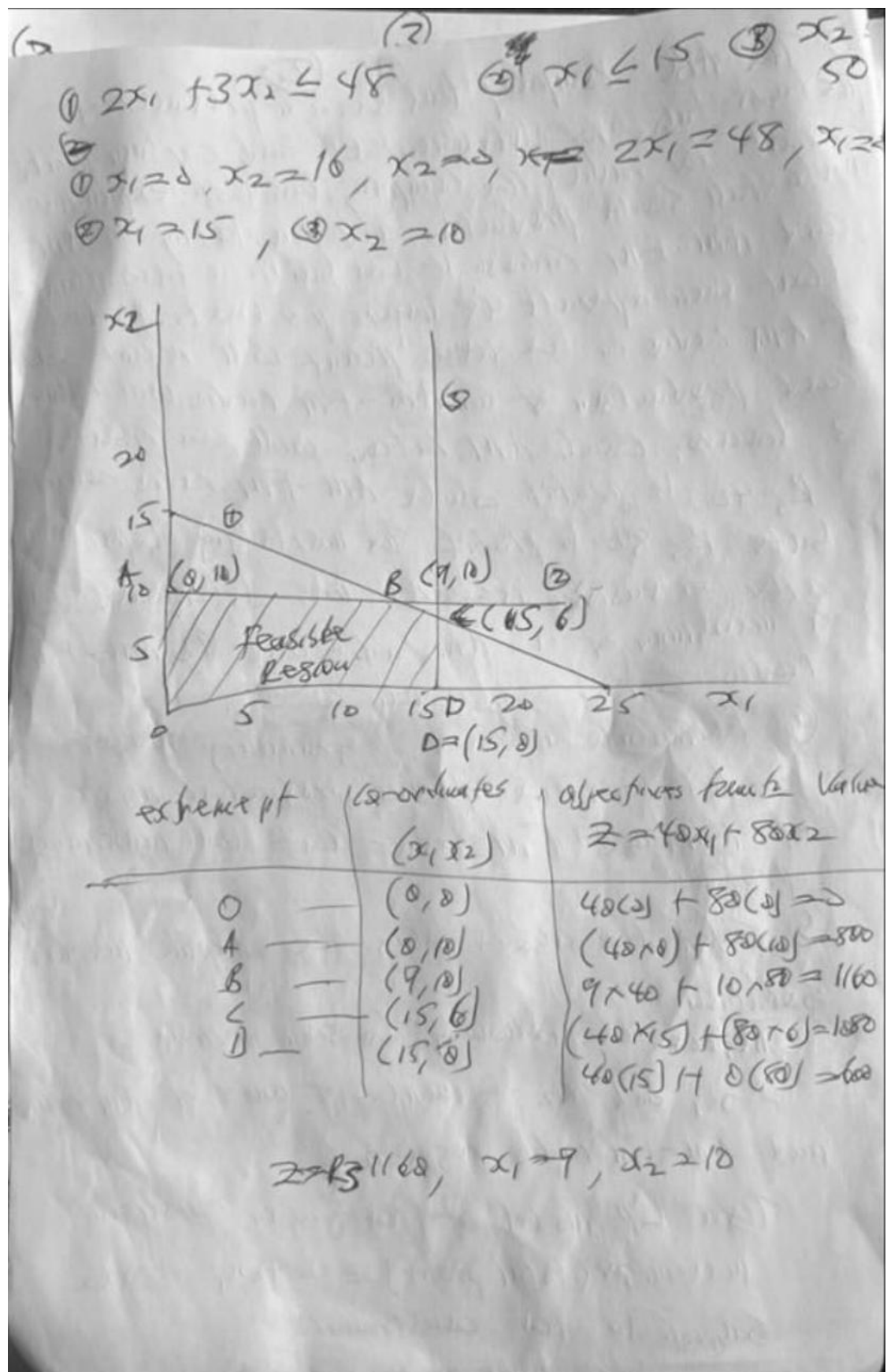
Solution
Define the following decision variable

x_1 and x_2 = number of units of AM radio and AM-FM radio respectively.

Then LP model of the given problem

maximize (total profit) $z = 40x_1 + 80x_2$

Subject to the constraints



(3) An electronics company produces two P_1 and P_2 products are produced and sold on a weekly basis. The weekly production cannot exceed 25 for product P_1 and 35 for product P_2 because of limited available facilities. The company employs total of 60 workers. Product P_1 requires 2 man-weeks of labour, while P_2 requires one man-week of labour. Profit margin on P_1 is Rs 60 and on P_2 is Rs 40. Formulate this problem and solve graphically for maximum profit.

Solution: Let define the following decision variables, x_1 and x_2 = number of units of product P_1 & P_2 respectively.

Then LP model of the given problem

$$\text{maximize } Z = 60x_1 + 40x_2$$

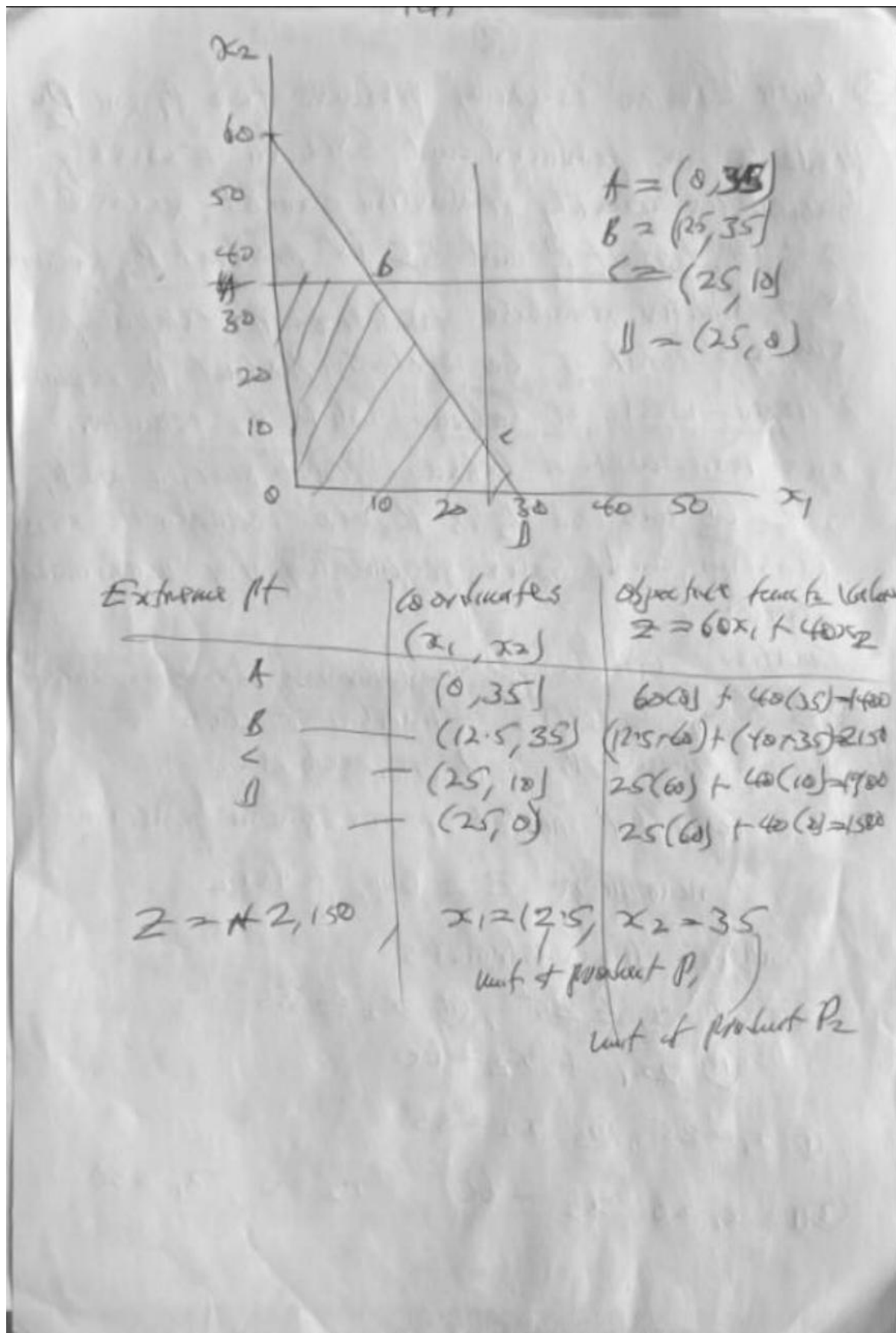
Subject to constraints

$$(i) x_1 \leq 25, (ii) x_2 \leq 35$$

$$(iii) 2x_1 + x_2 \leq 60$$

$$(iv) x_1 \geq 25 \quad (v) x_2 \geq 35$$

$$(3) (vi) x_1 \geq 0, x_2 \geq 0, x_1 \leq 30$$



④

Minimization LP Problem

G. J. Breweries Ltd have two bottling plants and located at "G" and the other at "J". Each plant produces three drinks, Whisky, Beer and Brandy named A, B and C respectively. The number of the bottles produced per day are shown in the table below.

Drink	Plant	
	G	J
Whisky	1500	1500
Beer	3000	1000
Brandy	2000	5000

A market survey indicates that during the month of July, there will be a demand of 20,000 bottles of Whisky, 40,000 bottles of Beer and 44,000 bottles of Brandy. The operating cost per day for plant at G and J are 600 and 400 monetary unit. For how days each plant be run in July so as to minimize the production cost, while still meeting the market demand? Solve graphically.

Solution:

Let us define the following decision variables

x_1 and x_2 = number of days of work.

Then the LP model of the given problem can be expressed as Minimize $Z = 600x_1 + 400x_2$

Subject to the constraints

$$\textcircled{1} 3x_1 + 3x_2 \geq 40 \quad \textcircled{2} 3x_1 + x_2 \geq 40$$

$$\textcircled{3} 2x_1 + 5x_2 \geq 44$$

$$\textcircled{1} 3x_1 + 3x_2 \geq 40$$

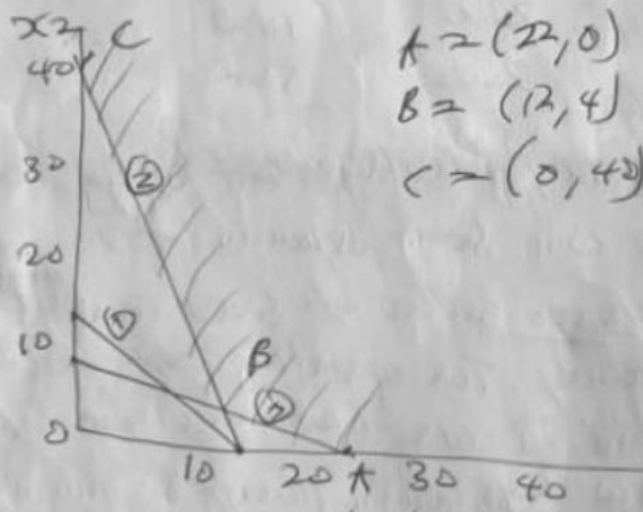
$$\textcircled{2} 3x_1 + x_2 \geq 40$$

$$\textcircled{3} 2x_1 + 5x_2 \geq 44$$

$$\textcircled{1} x_1 = 0, x_2 = 13.33, x_2 = 0, x_1 = 13.33$$

$$\textcircled{2} x_1 = 0, x_2 = 40, x_2 = 0, x_1 = 13.33$$

$$\textcircled{3} x_1 = 0, x_2 = 8.8, x_2 = 0, x_1 = 22$$



Extreme	Coordinate (x_1, x_2)	Objective function Value $Z = 600x_1 + 400x_2$
A	(22, 0)	$600(22) + 400(0) = 13,200$
B	(12, 4)	$600(12) + 400(4) = 8,800$
C	(0, 40)	$600(0) + 400(40) = 16,000$

The min-value of objective function occurs at point (12, 4) Plant C/min for $x_1 = 12$ day, Plant J $x_2 = 4$ day

To have a minimum production cost of Rs 8,800.
 ⑤ A pharmaceutical company has 100 kg of A, 180 kg of B and 120 kg of C ingredients available per month. The company can use these materials to make three basic pharmaceutical products namely S-10-5, S-5-10 and 20-5-10 where the numbers in each case represent the percentage of weight of A, B, and C respectively, in each of the products. The cost of these raw materials is as follows:

Ingredient	Cost per kg (Rs)
A	80
B	20
C	50
Cost ingredient	20

The selling price of these products are Rs 40.5, Rs 43 and Rs 45 per kg. respectively. There is a capacity restriction of the company for product S-10-5 less than 30 kg per month. Determine how much of each of the products the company should produce in order to maximize its monthly profits.

Solution :-

Let the P_1 , P_2 and P_3 be the three products to be manufactured. The data of the problem can be summarised as follows:-

(6) A company makes two kinds of leather belts A and belt B. Belt A is a high quality belt and belt B is of lower quality. The respective profit are Rs 4 and Rs 3 per belt. The production of each of type A requires twice as much time as a belt of type B and if all belt were of type B, the company could make 1000 belt per day. The supply of leather is sufficient for only 800 belt per day (both A and B combined). Belt A requires a fancy buckle and only 400 of these are available per day. There are only 700 buckles a day available for belt B. What should be the daily production of each type of belt? Formulate this problem as an LP model and solve it using Simplex method.

Solution

Let x_1 and x_2 be the no. of belt of type A & B respectively manufactured each day. Then the mathematical LP model would be written

$$\text{maximize (total profit) } Z = 4x_1 + 3x_2$$

Subjected to constraints:-

$$(1) \quad 2x_1 + x_2 \leq 1000 \quad (\text{time available})$$

$$(2) \quad x_1 + x_2 \leq 800 \quad (\text{supply of leather})$$

$$(3) \quad \begin{matrix} x_1 \leq 400 \\ x_2 \leq 700 \end{matrix} \quad \text{buckle availability}$$

$$x_1, x_2 \geq 0$$

$x_2 = 600$, $x_1 = 200$. Belts of A and $x_2 = 600$ belts of type B in order to obtain the maximum profit of Rs 2600.

⑦ ~~Wait~~ QUEUING THEORY

- ⑦ A television repairman finds that the time spent on his jobs has an exponential distribution with a mean of 30 minutes. If he repairs the set in order in which they come in and if the arrival of sets follows a poisson distribution with an approximate average rate of 10 per 8 hour day, what is the repairman's expected idle time each day? How many jobs are ahead of the average set just brought in.

Solution

From the data of the problem, we have

$$\lambda = 10/8 = 5/4 \text{ sets per hour}$$

$$\mu = 1/30 (60) = 2 \text{ set per hour}$$

- ④ Expected idle time of repairman each day
No. of hours for which the repairman remains busy in a 8-hour day (traffic intensity) is given by

$$8(\lambda/\mu) = 8(5/8) = 5 \text{ hours}$$

Hence, the idle time for a repairman in an 8 hour day will be $(8-5) = 3$ hours

- ⑤ Expected (or average) number of TV sets in the system

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{5/4}{2 - (5/4)} = 2 \text{ TV sets}$$

⑧ In a railway marshalling yard, goods trains arrive at a rate of 30 train per day. Assuming that the interarrival time follows an exponential distribution and the service time (time taken to handle a train) distribution is also exponential with an average of 36 minutes. Calculate:

- ⑨ expected queue size (line length)
 ⑩ probability that the queue size exceeds 10.
 If the input of train increases to an average of 33 per day, what will be the change in ⑨ and ⑩
 Solution.

$$\lambda = 30/60 \times 24 = 1/48 \text{ train per minute}$$

$$\mu = 1/36 \text{ train per minute}$$

$$\text{The traffic intensity then is } \rho = \frac{\lambda}{\mu} = \frac{36}{48}$$

- ⑨ Expected queue size (line length) = 0.75

$$L_s = \frac{\rho}{1-\rho} = \frac{0.75}{1-0.75} = 3 \text{ trains}$$

- ⑩ probability that the queue size exceeds 10

$$P(n \geq 10) = \rho^{10} = (0.75)^{10} = 0.06$$

If the input increase to 33 train per day, then we have

$$\lambda = 33/60 \times 24 = 11/480 \text{ train per minute}$$

$$\mu = 1/36 \text{ train per minute}$$

$$\text{Then intensity (traffic)} = \rho = \frac{\lambda}{\mu} = \frac{11}{480} \times 36 = 0.83$$

8

Hence recalculating the values for (a) & (ii)

$$L_s = \frac{\rho}{1-\rho} = \frac{0.83}{1-0.83} \approx 5 \text{ from approx.}$$

$$\text{and } P(n \geq 10) = \rho^{10} = (0.83)^{10} \approx 0.2 \text{ approx.}$$

(8) Arrivals at telephone booth are considered to be poisson with an average time 3 minutes b/w one arrival and the next. The length of phone calls is assumed to be distributed exponentially with a mean of 3 minutes.

(a) What is the probability that a person arriving at the booth will have to wait?

(5) The phone telephone department will install a second booth when convinced that an arrival would expect waiting for at least 3 minutes for a phone call. By how much should the flow of arrivals increase in order to justify a second booth?

(c) What is the average length of the queue that forms from time to time?

(d) What is the probability that it will take a customer more than 10 min altogether to wait for the phone and complete his call?

Solution

$$\lambda = 1/3 \approx 0.33 \text{ person per minute and } \mu = 1/3 \approx 0.33$$

(a) probability that a person has to wait at the booth

$$P(n > 0) = 1 - \rho = \frac{\lambda}{\mu} = \frac{0.1}{0.33} \approx 0.3$$

⑤ The installation of a second booth will be justified if the arrival rate is more than the waiting time. Let λ' be increased arrival rate. Then the expected waiting time in the queue will be

$$W_q = \frac{\lambda'}{\mu(\mu - \lambda')}$$

$$3 = \frac{\lambda'}{0.33(0.33 - \lambda')}$$

$$\lambda' = 0.16$$

Hence, the increase in the arrival rate is $0.16 - 0.18 = 0.06$ arrival per minute

⑥ Average length of non-empty queue.

$$L = \frac{\lambda}{\mu - \lambda} = \frac{0.33}{0.33 - 0.1} = \frac{0.33}{0.23} \approx 2$$

— Customers:

⑦ Probability of waiting for 10 minute or more is given by

$$P(t \geq 10) = \int_{10}^{\infty} \frac{\lambda}{\mu} (\mu - \lambda) e^{-(\mu - \lambda)t} dt$$

$$= \int_{10}^{\infty} (0.3)(0.23) e^{-0.23t} dt$$

$$= 0.069 \left[\frac{e^{-0.23t}}{-0.23} \right]_{10}^{\infty} = 0.03$$

This shows that on an average of 3% of arrival will have to wait for 10 min or more but they can use the phone.

- (9) A road transport company has one reservation clerk on duty at a time. He handles information & bus schedules and makes reservations. Customers arrival at a rate of 8 per hour and the clerk can, on an average, service 12 customers per hour. After stating your assumptions, answer the following

Solutions

It is given that $\lambda = 8$ per hour

$\mu = 12$ per hour

- (a) The average no. of customers waiting for service in the system are

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{8}{12 - 8} = 2 \text{ customers}$$

- (b) The average time spent by customers in the system before being served is

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{12 - 8} = \frac{1}{4} \text{ hour} = 15 \text{ minutes}$$

The average waiting time for a customer in the queue

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{8}{12(12 - 8)} = \frac{1}{6} \text{ hour} = 10 \text{ minutes}$$

- (10) Consider a single server queueing system with Poisson input and exponential service times. Suppose the mean arrival rate is 3 calling units per hour, the expected service time is 0.25 hours and the maximum permissible calling

units in the system is two. Derive the steady-state probability distribution of the number of calling units in the system and then calculate the expected number in the system.

Solution

$\lambda = 3$ unit per hour, $\mu = 4$ unit per hour

and $N = 2$

Then traffic intensity $\rho = \frac{\lambda}{\mu} = \frac{3}{4} = 0.75$

The steady-state probability distribution of the number of customers (calling units) in the sys is

$$p_n = \frac{(1-\rho)\rho^n}{1-\rho^{N+1}} = \frac{(1-0.75)(0.75)^n}{1-(0.75)^{2+1}} = (0.43)(0.75)^n$$

$\rho \neq 1$

$$p_0 = \frac{(1-\rho)}{1-\rho^{N+1}} = \frac{1-0.75}{1-(0.75)^{2+1}} = \frac{0.25}{1-(0.75)^3}$$

≈ 0.431

The expected number of calling units in the system is given by

$$L_s = \sum_{n=1}^N n p_n = \sum_{n=1}^2 n (0.43)(0.75)^n$$

$$= 0.43 \sum_{n=1}^2 n (0.75)^n = 0.43(0.75) + 2(0.75)^2$$

≈ 0.81

(11) A salesman located in a city A decided to travel to city B. He knows the distances of alternative routes from city B. He then drew a highway network map as shown below. The city of origin A is city 1. The destination city B is city 10. Other cities through which the salesman will have to pass through are 2 to 9. The arrows representing routes show cities and distances in kilometers are indicated on each route. The salesman's problem is to find the shortest route that covers all selected cities from A to B.

Soln

We start calculating distance by a pair of cities from destination city 10 ($=x_{10}$) and work backwards $x_5 \rightarrow x_4 \rightarrow x_3 \rightarrow x_2 \rightarrow x_1$ to find the optimal path. The recursive relationship for this problem can be stated as follows.

$$f_n^*(S_n) = \min_{d_n} \left[D_{S_n d_n} + \int_{n-1}^n (d_n) \right] \quad n=1,2,3,\dots$$

where $f_{n-1}^*(d_n)$ is the optimal distance for the previous stages.

We move backward to stage 3. Suppose that the salesman is at state $S_2 = 5$ (node 5). Then

We has to decide whether he should go to either $d_2 = 8$ (~~node 8~~) (node 8) or $d_2 = 9$ (node 9). For this he must evaluate from source

$$D_{S,8} + f_1^*(8) = 4 + 7 = 11 \quad (\text{to state } S_1 = 8)$$

$$D_{S,9} + f_1^*(9) = 8 + 9 = 17 \quad (\text{to state } S_1 = 9)$$

Decision $d_1 \rightarrow$		minimum distance $f_1^*(S_1)$	optimal decision d_1
State 1	8	7	10
	9	9	10

The distance function for travelling from $S_2 = 5$ is the smallest f_1^* plus $d_{S,5}$ from source.

$$f_2(S_2) = \min_{d_2=8,9} [11, 17] = 11 \quad (\text{to state } S_1 = 8)$$

Similarly the calculation of distance function for travelling from state $S_2 = 6$ or $S_2 = 7$ can be completed as follows.

For state $S_2 = 6$

$$f_2(6) = \min_{d_2=8,9} \begin{cases} D_{6,8} + f_1^*(8) = 3 + 7 = 10 \\ D_{6,9} + f_1^*(9) = 7 + 9 = 16 \end{cases} = 10 \quad (\text{to state } S_1 = 8)$$

For state, $s_2 = 7$

$$f_2(7) = \min_{d_2=8,9} \begin{cases} d_{18} + f_1^*(8) = 8 + 7 = 15 \\ d_{19} + f_1^*(9) = 4 + 9 = 13 \end{cases}$$

$$= 13 \text{ (to state } s_1 = 9)$$

These results are entered into the two stage table as shown in below

Decision $d_2 \rightarrow$	8	9	min. distance $f_2^*(s_2)$	optimal decision s_2
state s_2 5	11	17	11	8
6	10	16	10	8
7	15	13	13	9

The results that we obtain by continuing the same process for stage 4 & 5 are shown below

Decision $d_3 \rightarrow$	5	9	7	min. distance $f_3^*(s_3)$	optimal decision s_3
state s_3 2	18	20	18	18	5 or 7
3	14	18	17	14	5
4	17	20	18	17	5

Decision $d_4 \rightarrow$	2	3	4	min. dist $f_4^*(s_4)$	optimal decision s_4
state s_4 1	22	20	20	20	3 or 4

The above optimal results at various stages can be summarized as below

$$\text{Sequence} \begin{bmatrix} 10 & 8 & 5 & 3 & 1 \\ 10 & 8 & 5 & 4 & 1 \end{bmatrix}$$

$$\text{Distance} \begin{bmatrix} 7 & 4 & 3 & 6 = 20 \\ 7 & 4 & 6 & 3 = 20 \end{bmatrix}$$

From above, it is clear that there are two alternative shortest routes for this problem, both having a minimum distance of 20 kilometres.

Simulation & Inventory problems (17)

Theory & Application

- (12) A manager must choose b/w ^{two} investments A & B that are calculated to yield net profits of \$1200 and \$1600, respectively with probabilities subjectively estimated at 0.75 and 0.60. Assume the manager's utility function reveals that utilities for \$1200 and \$1600 are 45 and 50 units, respectively. What is the best choice on the basis of the expected utility value (EUV)?

Solution
The expected utility value is expressed as

$$EUV = \sum_{i=1}^n U_i P_i$$

U_i = utility value of state of nature i

P_i = probability value of state of nature i

$$EUV(A) = P_A U_A = 0.75 \times 45 = 33.75 \text{ unit}$$

$$EUV(B) = 0.6 \times 50 = 30 \text{ units}$$

Since $EUV(A) > EUV(B)$, Hence
the best choice is investment A.

(12)
A vegetable seller buys tomatoes for \$45 a box and sells them for \$80 per box. If the box is not sold ~~before~~ sold on the first selling day, it is worth \$15 as salvage. The past records indicate that demand is normally distributed with a mean of 30 boxes daily and a standard deviation of 9 boxes. How many boxes should be stock!

Solution

minimum required probability of selling at least an additional unit (box) to justify the stock at that unit is

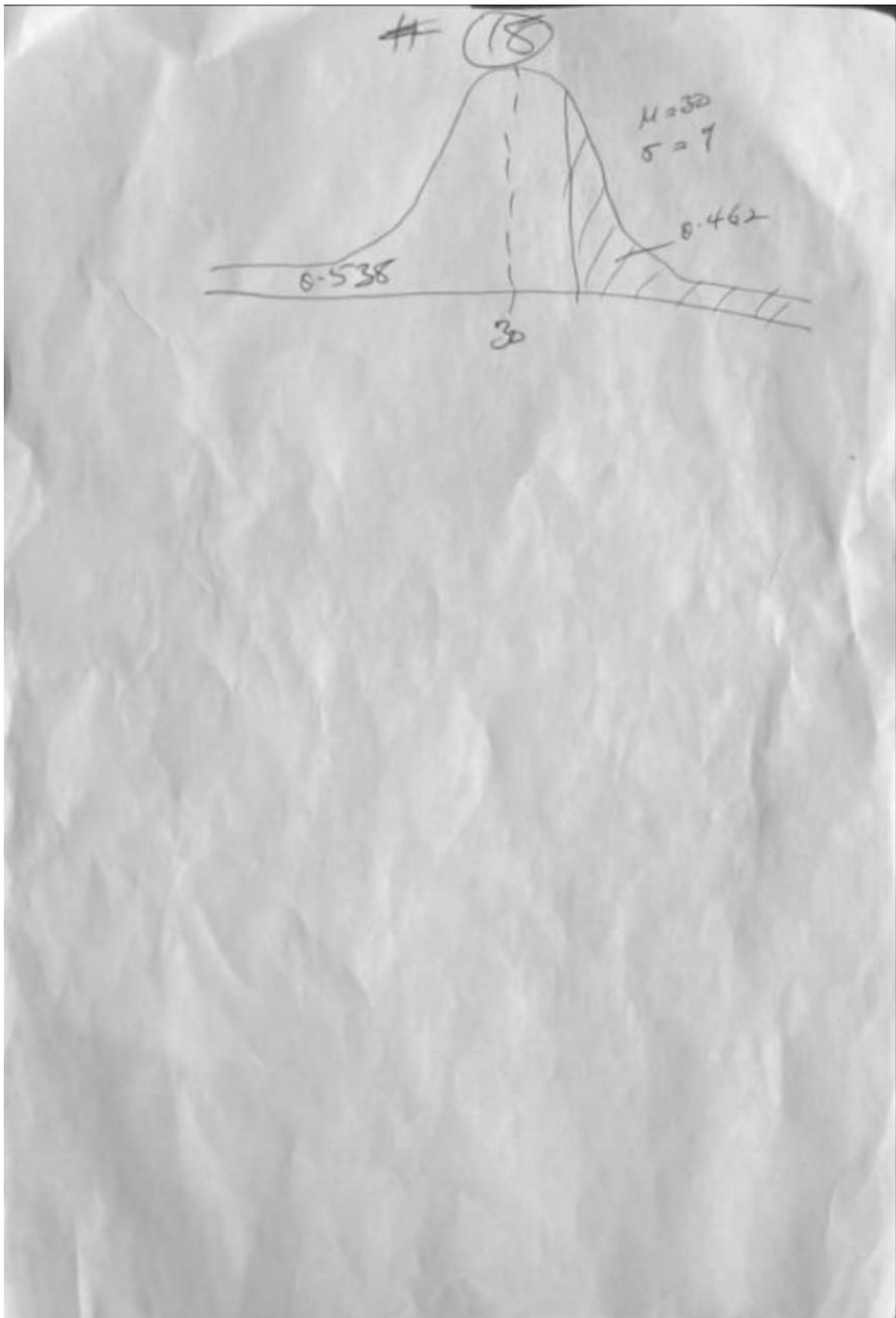
$$P = \frac{1L}{1P + 1L} = \frac{45 - 15}{(80 - 45) + (45 - 15)} = \frac{30}{45} = 0.462$$

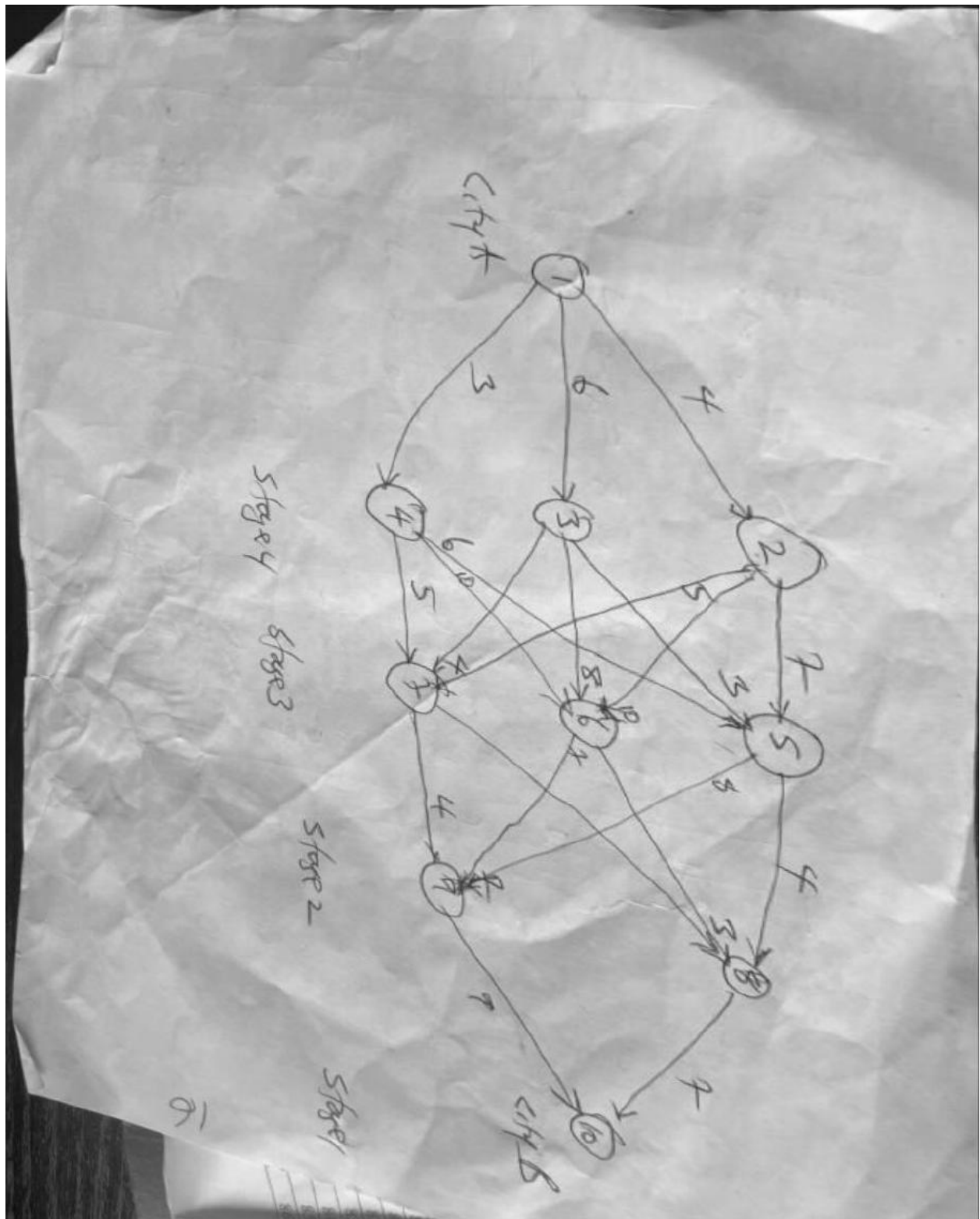
The vegetable seller must be 46.2% sure of selling at least an additional unit before it would pay him to stock that unit. The vegetable seller should stock additional boxes until he reaches point A. If he stocks more units, the probability will fall below 0.462.

This implies that pt. A is 0.1 standard deviation to the right of the $\bar{x} = 30$.

$$\begin{aligned} \text{Point A} &= \text{mean} + \text{standard deviation} \\ &= 30 + (0.1 \times 9) = 30.1 = 31 \text{ boxes} \end{aligned}$$

Hence the fruit seller should stock 31 boxes





Paper 2 of 2 — CHE 511 PROCESS OPTIMIZATION TECHNIQUES 2023/2024

Level 500 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

i

Tax os o2.G3"t

COLLEGE OF ENGINEERING AND TECHNOLOGY '

DEPARTMENT OF CHEMICAL ENGINEERING

FIRST SEMESTER; 2023/2024 SESSION EXAMINATION

COURSE TITLE: PROCESS OPTIMIZATION TECHNIQUES Credit units: 3

COURSE CODE: CHE \$11 Time Allowed: 3 Hours.

INSTRUCTIONS: Answer any five (5) Questions, All Questions carry equal marks

1.(a) The ABC Company has been a producer of picture tubes for television sets and certain printed circuits for radio. The company has just expanded into full scale production and marketing of AM and AM-FM radios. It has built a new plant that can operate 48 hours per week, Production of an AM radio in the new plant will require 2 hours and production of an AM-FM radio will required 3 hours. Each AM radio will contribute N40 to profit while an AM-FM radio will contribute N80 to profit. The marketing department, after extensive research has determined that a maximum of 15 AM radios and 10 AM-FM radios can be sold each week.

(i) Formulate a linear programming model to determine the optimum production mix of AM and AM-FM radios that will maximize profit.

(ii) Solve the above problem using the graphic method,

1.(b) A television repairman finds that the time spent on his jobs has exponential distribution with a mean of 30 minutes, If he repairs the set in order in which they come in and if the arrival of sets follows a poison distribution with an approximate average rate of 10 per 8 hour day. What is the repairman's expected idle time each day? How many jobs are ahead of the average set just brought in?

2. A company make two kinds of leather belts, belt A and belt B, Belt A is a high quality belt and felt B is of lower quality. The respective profit are N4 and N3 per belt, The production of each of type A requires twice as much time as a belt of type B and if all belt were of type B, the company could make 1000 belt per day. The supply of leather is sufficient for only 800 belt per day (both A and B combined). Belt A requires a fancy buckle and only 400 of these are available per day. There only 700 buckles a day available for belt B.

What should be the daily production of each type of belt? Formulate this problem as a linear problem madel and solve it using simplex method,

<a\

al

3A market gardener has a small plot of land she wishes to use to plant cabbages and tomatoes. From past experience she knows that her garden will yield about 1.5 tons of cabbage per acre or 2 tons of tomatoes per acre. Her experience also tells her that cabbage requires 20 lbs of fertilizer per acre, whereas tomatoes require 60 lbs of fertilizer per acre. She expects that her efforts will produce \$600 = ton of cabbages and \$750 = ton of tomatoes. Unfortunately, the truck she uses to take her produce to market is getting old, so she doesn't think that it will transport any more than 3 tons of produce to market at harvest time. She has also found that she only has 60 lbs fertilizer. What combination of cabbages and tomatoes should she plant to maximize her products. Using simplex method.

4, Write in details the Optimization of a Chemical reactor.

Hints (Optimization in design and operation of a reactor focuses on formulating a suitable objective function plus a mathematical description of the reactor).

5. (a) Write briefly on Trajectory, also state relevant /supporting equations.

(b) Modelling velocity from Newton's second law. Dd

(c) Modelling rotation.

(d) Briefly explain Runge-Kutta method, in determining the angle and velocity of a moving Rocket.

6/(a) A road transport company has one reservation clerk on duty at a time. He handles information of bus schedules and makes reservations. Customer's arrival at a rate of eight per hour and the clerk can, on an average, service 12 customers per hour. After stating your assumption, answer the following;

(1) The average number of Customers waiting for service in the system.

(2) The average time spent by Customer in the system before being served.

(b) Write briefly on four (4) assumptions of Linear Programming Model..

(c) Write briefly on four advantages of using linear Programming.

(d) Write briefly on four (4) limitations of Linear Programming.

7. (a) A salesman located in a city A decided to travel to city B. He knows the distances of alternative routes from city B. He then drew a highway network map as shown below. The city of origin A is city 1. The destination city B, is city 10. Other cities through which the salesman will have to pass through are 2 to 9. The arrow representing routes between cities and distances in kilometers are indicated on each route. The salesman's problem is to find the shortest route that covers all selected cities from A to B.

HIGHWAY NETWORK MAP FOR QUESTION 7

(b) State four (4) advantages of Simulation.

(c) State four (4) disadvantages of Simulation,

(d) List four (4) types of Simulation.